

Def. Let  $A \in M_{m \times n}(F)$  be a  $m$  by  $n$  matrix

The first elementary Row operation is: for any transposition  $(i \ j) \in S_m$ ,

$$E_{i,j}^{1st} \stackrel{\text{def}}{=} \begin{bmatrix} e_{(i,j)(1)} \\ e_{(i,j)(2)} \\ \vdots \\ e_{(i,j)(m)} \end{bmatrix} \quad (E_{i,j}^{1st} A \text{ is a } i\text{th and } j\text{th row-changed matrix.})$$

The second elementary Row operation is: for any non-zero scalar  $c \in F$ , and  $k \in \{1, \dots, m\}$

$$E_c^{2nd} \stackrel{\text{def}}{=} \begin{bmatrix} e_1 \\ \vdots \\ c \cdot e_k \\ \vdots \\ e_m \end{bmatrix} \quad (E_c^{2nd} \cdot A \text{ is multiplying } c \text{ to } k\text{th row})$$

The third elementary Row operation is: for any scalar  $c \in F$  and  $1 \leq i \neq j \leq m$ ,

$$E_{c,i,j}^{3rd} \stackrel{\text{def}}{=} \begin{bmatrix} e_1 \\ \vdots \\ c \cdot e_i + e_j \\ \vdots \\ e_m \end{bmatrix} \leftarrow j\text{th column} \quad (E_{c,i,j}^{3rd} A \text{ is } c \cdot (i\text{th row}) \text{ adding to } j\text{th row.})$$

Moreover, these are all invertible,

$$(E_{i,j}^{1st})^{-1} = E_{i,j}^{1st}, \quad (E_c^{2nd})^{-1} = E_{c^{-1}}^{2nd}, \quad (E_{c,i,j}^{3rd})^{-1} = E_{-c,i,j}^{3rd}$$

Rank of a Matrix.

Def. Given a Matrix  $A \in M_{m \times n}(F)$ , define

$$\text{rank } A \stackrel{\text{def}}{=} \dim \text{Im } L_A.$$

Clearly,  $\text{rank } A \leq m$ .

Prop. Let  $A \in M_{m \times n}$ , and  $P \in M_{m \times m}$  and  $Q \in M_{n \times n}$  be invertibles, Then

$$\text{rank } PAQ = \text{rank } A.$$

Proof. Observe that

$$\text{Im } L_{PAQ} = \text{Im}(L_P \circ L_A \circ L_Q) = (L_P \circ L_A \circ L_Q)[F^n] = (L_P \circ L_A) \underbrace{[F^n]}_{L_Q \text{ onto}} = \text{Im } L_A.$$

Moreover, since  $L_P$  is an Isomorphism,  $L_P[L_A[F^n]] \cong L_A[F^n] = \text{Im } L_A$ .  $\square$

Thm. Given a Matrix  $A \in M_{m \times n}$ ,

$$\text{rank } A = \dim \text{span}\{a_1, \dots, a_n\} \text{ where } a_i \text{ is the } i\text{th column of } A. \\ (\text{or } a_i = A e_i)$$

Proof.  $\text{rank } A = \text{rank } L_A = \dim \text{Im } L_A = \dim \text{span}\{L_A(e_1), \dots, L_A(e_n)\} = \dim \text{span}\{a_1, \dots, a_n\}.$

Thm. Given a Matrix  $A \in M_{m \times n}$  of rank  $A = r$ , then there are invertibles  $B \in M_{m \times m}$ ,  $C \in M_{n \times n}$ ,

$$\text{such that } BAC = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix}$$

Thm. Let  $T: V \rightarrow W$  and  $U: W \rightarrow Z$  be linear maps on fin. dim  $V, W, Z$ . Then

$$\text{rank } U \circ T \leq \text{rank } U \text{ and } \text{rank } U \circ T \leq \text{rank } T.$$

Proof.  $\text{Im } U \circ T = (U \circ T)[V] \subseteq U[W] = \text{Im } U$ , so  $\text{rank}(U \circ T) \leq \text{rank } U$ .

$$\text{So, } \text{rank}(U \circ T) = \text{rank}(T^t \circ U^t) \leq \text{rank } T^t = \text{rank } T \quad \square$$

Def. Given  $A \in M_{m \times n}(F)$ , Consider the linear map

$$L_A: F^n \rightarrow F^m; X \mapsto AX$$

Then,  $\ker L_A$  is the solution set of the homogeneous linear system  $AX=0$ .

Thm. For any kind elementary Row operation  $E$ ,

$$\ker L_A = \ker L_{EA}.$$

Proof.  $AX=0$ , then clearly  $EAX = E0 = 0$ . So  $\ker L_A \subseteq \ker L_{EA}$ .

Conversely, since  $E$  is invertible,  $EAX=0 \Rightarrow AX=E^{-1}0=0$ .

Thm. Given a  $m$  by  $n$  matrix  $A \in M_{m \times n}(F)$  and  $B \in F^m$ ,

if  $S \in F^n$  satisfies  $AS=B$ , then

$$x \in F^n \text{ satisfies } Ax=B \Leftrightarrow x \in S + \ker L_A.$$

Proof. If  $Ax=B$ , then  $Ax=B=AS$ , so  $A(x-S)=0$ . Thus  $x-S \in \ker L_A$ , or  $x \in S + \ker L_A$ .

Conversely,  $x \in S + \ker L_A$  be given. That is,  $x=S+\gamma$  for some  $\gamma \in \ker L_A$ , so

$$Ax = A(S+\gamma) = B + 0 = B.$$

Corollary.  $Ax=b$  has only one solution iff  $A$  is invertible.

Corollary. Given linear system  $Ax=b$  where  $A \in M_{m \times n}$ ,  $x \in F^n$ ,  $b \in F^m$ ,

$Ax=b$  has a solution if and only if  $\text{rank } A = \text{rank } (A|b)$ .

Proof. Since  $Ax=b$  for some  $x \in F^n$  if and only if  $b \in \text{Im } A$ , and

$$\text{Im } A = \text{Span}\{Ae_1, \dots, Ae_n\}$$

So, proved.  $\square$

In Inner Product space, we can use inner product to find "best" approximation solution.

Thm. Let  $A \in M_{m \times n}(\mathbb{C})$  and  $y \in \mathbb{C}^m$  be given.

Then, there exists  $x_0 \in \mathbb{C}^n$  such that

$$A^* A x_0 = A^* y \quad \text{and} \quad \|A x_0 - y\| \leq \|A x - y\| \text{ for all } x \in \mathbb{C}^n.$$

In particular, if  $\text{rank } A = n$ , then  $x_0 = (A^* A)^{-1} A^* y$ .

Proof. Given  $y \in \mathbb{C}^m$ , there exists an orthogonal projection  $A x_0$  on  $\text{Im } A$ .

That is,  $\langle A x_0 - y, A x \rangle = 0$  for all  $x \in \mathbb{C}^n$ .

And, since  $\langle A x_0 - y, A x \rangle = \langle A^* A x_0 - A^* y, x \rangle$ , so  $A^* A x_0 - A^* y = 0$ .

(Specifically, for orthonormal basis  $\{\alpha_1, \dots, \alpha_n\}$  for  $\text{Im } A \subseteq \mathbb{C}^m$ ,

$$A x_0 = \sum_{i=1}^n \langle y, \alpha_i \rangle \cdot \alpha_i.)$$

Moreover, if  $\text{rank } A = n$ , then  $\text{rank } A^* A = n$ , so invertible.